

Hands-on Lab: Getting Started with GitHub

Estimated time: 20 min

In this lab, you will get started with GitHub by creating a GitHub account and project and adding a file to it using its Web interface.

Objectives

After completing this lab, you will be able to:

1. Describe GitHub
2. Create a GitHub account
3. Add a project and repo
4. Edit and create a file
5. Upload a file and Commit

GitHub Overview

First, let's introduce you to GitHub. GitHub is a collection of folders and files. It is a Git repository hosting service, but it adds many of its own features. Git is a command-line tool. It hosts and maintains a server via command line. GitHub provides this Git server and a Web-based graphical interface for you. It also provides access control and collaboration features, such as wikis and basic task management tools for every project. In addition, GitHub provides cloud storage for source code, supports all popular programming languages, and streamlines the iteration process. GitHub includes a free plan for individual developers and hosting Open Source projects.

Exercise 1: Creating a GitHub Account

Please use the following steps to create an account on GitHub:

Step 1: Create an account: <https://github.com/join>

Sign up to GitHub

Email

Password

Username

Continue >

By creating an account, you agree to the [Terms of Service](#). For more information about GitHub's privacy practices, see the [GitHub Privacy Statement](#). We'll occasionally send you account-related emails.

NOTE: If you already have a GitHub account, you can skip this step and simply log in to your account.

Step 2: Provide the necessary details to create an account as shown below:

Sign up to GitHub

Email

Password

Username

Continue >

By creating an account, you agree to the [Terms of Service](#). For more information about GitHub's privacy practices, see the [GitHub Privacy Statement](#). We'll occasionally send you account-related emails.

Step 3: Click [Visual puzzle](#) to verify the account.

Verify your account

Please solve a puzzle so we can safely create your account.

Visual puzzle

Audio puzzle

Step 4: Solve the puzzle and [Submit](#).

Verify your account

Use the arrows to rotate the object to face in the direction of the hand. (1 of 1)

Match this angle

Submit

Audio puzzle

Restart

Step 5: Open your email, find the GitHub verification email, copy the verification code, and return to the GitHub Signup page to enter it.

Step 6: Confirm your email address using the verification code and [Continue](#).

Confirm your email address

We have sent a code to @gmail.com

Enter code

Continue >

Didn't get your email? [Resend the code](#) or [update your email address](#).

NOTE: If you do not receive the verification email, click [Resend the code](#).

Step 7: Sign in to GitHub, enter Id and Password and [Sign in](#).

Sign in to GitHub

Your account was created successfully. Please sign in to continue

Username or email address

Password

[Forgot password?](#)

Sign in

[Sign in with a passkey](#)

New to GitHub? [Create an account](#)

You will see the home page.

Home

What are Python developers?

Python Panda data analysis

Full authentication endpoint

Learn with a tutorial project

Introduction to GitHub

GitHub Pages

Code with Copilot

Start writing code

Start a new repository for pandasipatt

Introduce yourself with a profile README

Explore repositories

bagisto / bagisto

erere / wstunnel

Homebrew / homebrew-cask

Explore repositories

bagisto / bagisto

erere / wstunnel

Homebrew / homebrew-cask

Exercise 2: Adding a project and repo

Step 1: Click the [+](#) symbol and click [New repository](#).

New repository

Import repository

Explore repositories

bagisto / bagisto

erere / wstunnel

Homebrew / homebrew-cask

Create a new repository

Repository name *

TestRepo

Description (optional)

Testing repository

Public

Initialize this repository with:

Add a README file

Add .gitignore

Choose a license

Create repository

Step 2: Provide a name for the repository and initialize it with the empty [README.md](#) file.

Create a new repository

A repository contains all of your project's files, revision history, and collaborative discussion.

Required fields are marked with an asterisk (*)

Owner *

Repository name *

TestRepo

Great repository names are short and memorable. Need inspiration? How about [automatic-chainsaw](#)?

Description (optional)

Testing repository

Public

Private

Initialize this repository with:

Add a README file

Add .gitignore

Choose a license

Create repository

Click [Create repository](#).

Now, you will be redirected to the repository you have created.

Let us start editing the repository.

Exercise 3: Create and edit a file

Exercise 3a: Edit a file

Step 1: Once the repository is created, the root folder of your repository is listed by default, and has just one file, [README.md](#). Click the pencil icon to edit the file.

TestRepo

main

1 branch

0 tags

Go to file

Add file

Code

patelppojatt1

Initial commit

3 minutes ago

1 commit

README.md

Initial commit

3 minutes ago

TestRepo

Testing repository

Step 2: Add some text to the file.

patelppojatt1 / TestRepo

Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

TestRepo / README.md

in main

Edit

Preview

Code 55% faster with GitHub Copilot

1

TestRepo

2

Testing repository

3

This is the first markdown file.

Step 3: After adding the text and click [Commit changes](#).

Commit changes

Commit message

Update README.md

Extended description

Add an optional extended description...

Commit directly to the main branch

Commit a new branch for this commit and start a pull request

Commit changes

Now, check that your file is edited with the new text.

Exercise 3b: Create a new file

Step 1: Click the repository name to return to the master branch, like in this testrepo.

TestRepo

main

1 branch

0 tags

Go to file

Add file

Code

patelppojatt1

Update README.md

17 minutes ago

1 commit

README.md

Update README.md

17 minutes ago

TestRepo

Testing repository

Step 2: Click [Add file](#) and select [Create new file](#) to create a file in the repository.

TestRepo

main

1 branch

0 tags

Go to file

Add file

Code

patelppojatt1

Create firstpython.py

55 min

1 commit

README.md

Update README.md

17 minutes ago

TestRepo

Testing repository

Step 3: Provide the file name and the extension of the file. For example, firstpython.py and add the lines.

TestRepo

main

1 branch

0 tags

Go to file

Add file

Code

patelppojatt1

Add firstpython.py via upload

17 minutes ago

1 commit

README.md

Update README.md

17 minutes ago

TestRepo

Testing repository

Step 4: Now, your file is uploaded in the repository.

TestRepo

main

1 branch

0 tags

Go to file

Add file

Code

patelppojatt1

Add firstpython.py via upload

17 minutes ago

1 commit

README.md

Update README.md

17 minutes ago

TestRepo

Testing repository

Summary

In this document, you have learned how to create a new repository, add a new file, edit a file, upload a file in a repository, and commit the changes.

Author(s)

Romeo Kienzler

Malika Singla

Other Contributor(s)

Rav Ahuja

Skills Network

IBM